

Virtualization of Industrial Control System Testbeds for Cybersecurity

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ABSTRACT

With an immense number of threats pouring in from nation states and hacktivists as well as terrorists and cybercriminals, the requirement of a globally secure infrastructure becomes a major obligation. Most critical infrastructures were primarily designed to work isolated from the normal communication network, but due to the advent of the “Smart Grid” that uses advanced and intelligent approaches to control critical infrastructure, it is necessary for these cyber-physical systems to have access to the communication system. Consequently, such critical systems have become prime targets; hence security of critical infrastructure is currently one of the most challenging research problems. Performing an extensive security analysis involving experiments with cyber-attacks on a live industrial control system (ICS) is not possible. Therefore, researchers generally resort to test beds and complex simulations to answer questions related to SCADA systems. Since all conclusions are drawn from the test bed, it is necessary to perform validation against a physical model. This paper examines the fidelity of a virtual SCADA testbed to a physical test bed and allows for the study of the effects of cyber-attacks on both of the systems.

Keywords

PLC, SCADA, ICS, Cybersecurity, Vulnerability Analysis, OpenPLC, MODBUS, Simulink.

1. INTRODUCTION

A supervisory control and data acquisition (SCADA) system is used to control industrial systems. Major components of a SCADA system include a human machine interface (HMI), which allows a user to monitor and manage the system, and also programmable logic controllers (PLCs) which serve to control certain aspects of the physical infrastructure, such as sensors and actuators. Critical infrastructures are assets that are essential for the functioning of modern society. Some examples of critical infrastructure include the power grid, nuclear power plants, oil refineries, and chemical plants. Any equipment failure or act of a cyber-attack on such infrastructure can result in catastrophic consequences.

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In 1996 around 957,000 gallons of oil was spilled in a nearby river in Fork Shoals, South Carolina, USA when an operator acted on erroneous data [1]. Similar incidents can happen if an attacker compromises a SCADA system. A typical man-in-the-middle attack can be used to alter the data shown on the HMI to confuse operators to make bad operational decisions. In August 2003, one of the largest blackouts in America's history might have been caused by a cyber-attack. Around 50 million people were affected [2]. A nuclear power plant in Oak Harbour, Ohio, USA was compromised by the Slammer SQL server worm. The worm managed to disable the safety monitoring system of the power plant for nearly five hours [3]. Recent attacks involving STUXNET [4] and HAVEX [5] have shown how fatal vulnerabilities can prove to be. Public communities discussing SCADA vulnerabilities like scadastrangelove.org [6] have identified 150 zero day vulnerabilities in SCADA systems. Given these examples where critical infrastructure was compromised, it can be seen that research in SCADA security is very crucial. It is important to understand which impacts an attack can have on critical infrastructure and how to protect against them.

However, not much work has been done to validate the effect of the attacks on virtual and physical SCADA systems to ensure that the virtual system accurately represents a real system. Therefore, the goal of this project is to create two testbeds - one physical and one virtual, where the virtual is a model of the physical. It is also desired to perform a comprehensive study on how the test beds will behave in normal operational conditions and during a cyber-attack. If there is a high degree of fidelity between the two, the behavior of the virtual testbed is expected to be very similar to the physical one.

2. RELATED WORKS

To better facilitate this sort of research, people have found it useful to create test beds for critical infrastructure. Queiroz et. al. proposes a modular SCADA testbed where real sensors can be interfaced using the Modbus/TCP protocol [7]. Farooqui et. al. puts forth an inexpensive approach of developing SCADA test beds using Trutime, a MATLAB/Simulink based tool [8]. Adhikari et. al. presents a wide area measurement system cyber-physical test bed using real time digital simulator and hardware-in-the-loop. Even an automated simulation and control engine was developed to randomize faults and cyber-attacks [9]. These testbeds can be physical, virtual or both. In the case of virtual testbeds, they are essentially computer simulations. An example of a virtual testbed for studying cyberattacks on the power grid was created by researchers at Texas A&M. DoS attacks and Man-in-the-Middle attacks were developed for that testbed [10].

3. GAS PIPELINE TESTBED

Gas pipelines are important infrastructures, not only for the power generation sector, but also for other sectors such as petrochemicals, chemicals, transportation, manufacturing, and district heating. The United States Patriot Act defines critical infrastructure as “systems and assets, whether physical or virtual, so vital to the United States that the incapacity or destruction of such systems and assets would have a debilitating impact on security, national economic security, national public health or safety, or any combination of these matters.” [11] By this definition, gas pipelines can be considered critical infrastructures, and therefore deserve special attention in terms of cyber-security.

The Gas Pipeline Testbed is a small closed pipeline that tries to mimic the behavior of a real gas pipeline. An electrical pump increases the pressure inside the pipeline by pumping air into it. There are also pressure sensors constantly measuring the pressure inside the pipeline. A solenoid valve alleviates the internal pressure when opened. The pump, the solenoid valve and the digital pressure sensor are connected to a Programmable Logic Controller (PLC) that controls the system. The role for the PLC is to maintain pressure between a high and a low setpoint by turning on and off the pump. There is also a manual operation mode that enables the user to manually control the pump and the valve from the HMI.

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4. VIRTUALIZATION OF SCADA COMPONENTS

Any SCADA system can usually be divided into five components: physical system, wire bridge, PLC, SCADA Protocol and Human-Machine Interface. Fig. 1 shows how the individual parts are correlated. By segmenting the system, it is possible to virtualize each part of it and compare fidelity between the physical and the virtual counterpart of each individual segment. Also, the virtual and physical segments are interchangeable, which means that the system can still operate with some pieces virtualized.

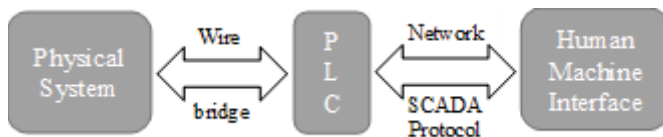


Figure 1. SCADA Components

4.1 Physical System

The physical system of the testbed is a gas pipeline connected to a pump and a pressure relief valve. The pump is a positive displacement pump connected to a 0.5 Hp 1Φ 120 Volts induction motor. The gas pipeline has a one inch diameter pipeline network, with four 90° pipe bends and two T-joints.

The virtual counterpart of the physical system was modeled in Simulink/Matlab. SimHydraulics was extensively used along with other toolboxes to replicate the physical characteristics of the

system. Fig. 2 compares the normal behavior of the physical and the virtual systems operating between 20 and 30 PSI.

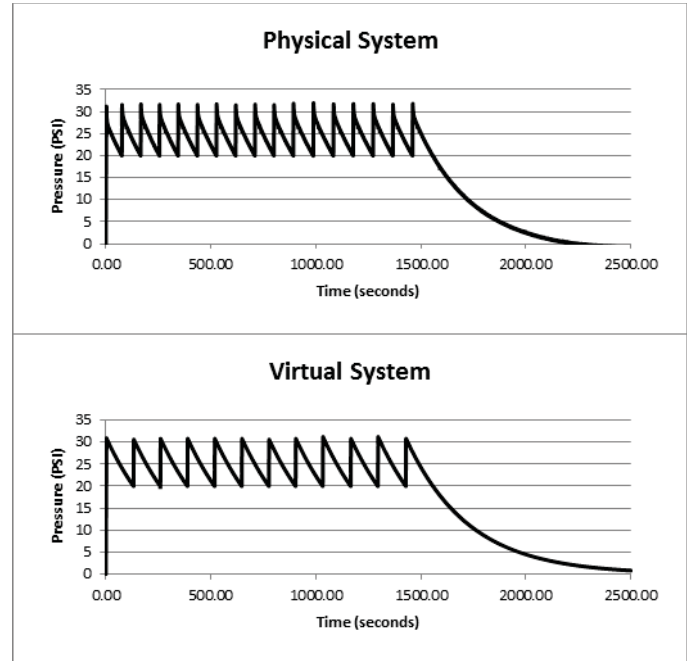


Figure 2. Pressure increase and decrease comparison for the virtual and physical systems

The small difference between both curves is mostly because of the linear rise of pressure in the virtual model. Matlab does not model the back pressure scenario that arises whenever any gas or liquid is pumped into an enclosed environment. The back pressure reduces the net output head of a pump, and hence the rise of pressure is never a linear phenomenon.

4.2 Wire bridge

Wires are the bridge between the physical system and the PLC. They are the way for which the PLC can read the data coming from the sensors and control the actuators based on its internal program.

In the virtual environment, the connection between the PLC and the Simulink model is made by network UDP packets that act as virtual wires for the system. An interface program was written in order to capture the data from the Matlab model and send them to the PLC. A virtual I/O driver was written for the PLC to interpret these packets as if they were local I/O.

Although the physical PLC is constantly connected by wires to the elements of the gas pipeline, the values are sampled at constant intervals, matching its scan time. The virtual I/O driver was written to match the same interval in order to increase fidelity with the physical testbed.

4.3 PLC

The PLC is an industrial digital computer that constantly monitors the state of its input lines (sensors) and makes decisions on what to do with its output lines (actuators) based on a custom program. It is one of the most important pieces in a SCADA system.

Since the goal for this project is to create a physical and a virtual testbed, it was important to choose a PLC that could control both.

The OpenPLC [12][13] is an open source soft-PLC, created to be an affordable alternative for automation and is also a valuable research tool as its entire source code is available. Because it is a soft-PLC, it can run on different hardware platforms, instead of a custom hardware. It supports all five IEC-61131-3 programming languages: Ladder Logic, Structured Text, Instruction List, Function Block Diagram and Sequential Function Chart. This makes the OpenPLC compatible with programs written for other standardized IEC-61131-3 PLCs. Also, the OpenPLC communicates with supervisory software using Modbus/TCP, one of the most common SCADA protocols [14].

On the physical testbed, the OpenPLC runs on a UniPi board [15]. The UniPi is an extension board for the RaspberryPi minicomputer. It has 8 changeover relays, 14 galvanically isolated digital inputs, 2 analog inputs (0-10V) and 1 analog output (0-10V). The virtual testbed has the OpenPLC running on Linux in a Virtual Machine. This is a similar environment to the RaspberryPi minicomputer.

The OpenPLC has a scan time of 50ms, which means that every 50ms it must read all inputs, scan through the entire program and perform all the output operations.

One downfall of using the OpenPLC in a virtualized environment is the non-real-time behavior caused by the virtualization layer. Table 1 shows the comparison of the real-time performance between the virtual and the physical OpenPLC over a time period of 16 minutes. It can be observed that the physical OpenPLC maintains a constant cycle time of 50ms with a very small standard deviation.

Table 1. Real-time response of the physical and virtual OpenPLC

PLC Cycle	Physical OpenPLC (ms)	Virtual OpenPLC (ms)
Processing (average)	2.84	0.00292
Sleeping (average)	47.16	49.99705
Total (average)	50	50
Standard Deviation	0.02089	0.08823
Max Deviation	0.31674	4.44

The virtual OpenPLC processes the logic much faster than the physical one because it is running on a more powerful hardware. However, due to the extra layer of virtualization, it has an increase of 23.68% on the standard deviation of the cycle time, with a 4.44ms max deviation, which means that it can't always wake up at the right time. It happens because while the OpenPLC's main thread sleeps on the VM, the Host OS may schedule the VM process out, therefore the OpenPLC has no control about when it must wake up the main thread again. When the Host OS finally schedules the VM process back, it might be too late to wake up the main thread in time. This core difference between the two systems can cause a different behavior in hard real-time applications. However, for slow-paced process like a gas pipeline, this doesn't interfere in the results.

4.4 Network / SCADA Protocol

The Modbus protocol is one of the most common SCADA protocols in use and is also supported by the OpenPLC. Therefore, MODBUS/TCP was selected to be the SCADA protocol for the Gas Pipeline Testbed.

Both the virtual and the physical OpenPLC use network interfaces to perform this communication. However, since one Network Interface Card (NIC) is emulated through the Virtual Machine Manager (VMM) software and the other is the physical card present in the RaspberryPi microcomputer, they might have different response times for handling network packets.

4.5 Human Machine Interface (HMI)

On a SCADA system, the HMI is a software application that graphically presents information about the current process state and also allows the operator to control some elements of the process. In order to display the most updated information, the HMI must constantly fetch data from the PLC.

For the Gas Pipeline Testbed, the HMI was created in C# and uses the EasyModbusTCP library [16] to communicate with the OpenPLC over Modbus/TCP. It queries the OpenPLC for new data every 100ms, and then updates the digital pressure meter to reflect any pressure change in pound per square inch (psi) in the gas pipeline. The system can either operate in manual mode where the user will have full control over the pipeline or it can be automated by putting the system in pump mode. In pump mode the OpenPLC keeps the gas pipeline in between two given set points. The Status of the pump is shown on the HMI by red or green color which signifies OFF or ON respectively.

Since both of the testbeds uses the same controller and SCADA protocol, the same HMI works with both the virtual and the physical testbeds. It is expected that the response of the HMI would be identical to the requests made by the Supervisory Control Software and would provide a fairer comparison between the two testbeds.

5. TESTBED BEHAVIOR DURING CYBER ATTACKS

The attacks that have been implemented in this work can be classified as Denial-of-Service (DoS) and Man-in-The-Middle (MiTM) attacks. The same experiments were conducted on both the Physical testbed and the Virtual testbed to observe the resulting behavior.

5.1 DoS Attack

A DoS attack is essentially an attack in which communication is prevented between nodes in a network [17]. There are multiple ways a DoS attack can be achieved. One of them is called volumetric DoS, in which the target is flooded with network packets so that the real communication is injured.

A volumetric DoS attack was performed using hping3, targeting the OpenPLC controller. When flooded with packets from hping3, the physical OpenPLC not only stopped responding to the HMI but also began to control the testbed with a large amount of lag.

A comparison between the real-time response of the physical and the virtual OpenPLC during the DoS attack can be seen in Figure 3. The physical OpenPLC turned on the pump when the pressure got below the setpoint. However, as the system started to get

overwhelmed by the flood packets, the OpenPLC response starts to deviate from the general 50ms. Finally, during the 280-300 seconds instance, the turnoff operation got delayed for staggering 12 seconds. In order to prevent overpressure, the emergency valve of the gas pipeline had to be opened.

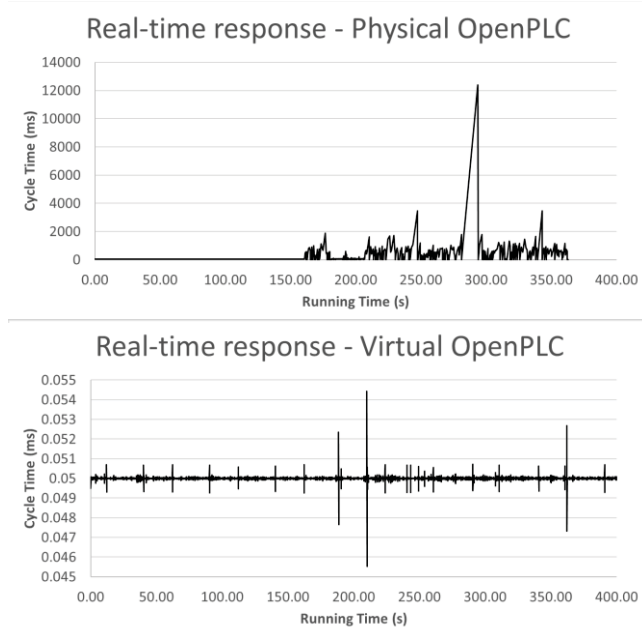


Figure 3. Real-time response comparison during DoS attack

Due to the differences in processing power between the virtual and the physical testbeds, similar behavior was not shown by the virtual OpenPLC as it remained unaffected by the volumetric DoS attack.

5.2 MiTM Attack

A man-in-the-middle (MITM) attack is a cyber-attack in which a malicious node is able to intercept the communications between two nodes. The attacker is able to see the data it intercepts and also modify it [18]. Two different MiTM attacks were successfully implemented on both the virtual and the physical testbed using ARP poisoning. Ettercap was used to perform the MiTM and custom filters were written in etterfilter to drop or modify the data.

The first MiTM attack caused a denial-of-service on the HMI as it dropped all Modbus packets between the OpenPLC and the HMI. Therefore, the HMI did not receive any real-time information about the system that it was monitoring and lost the ability to change control parameters. However, the behavior of the system was not affected, as the OpenPLC continued to operate normally. Both virtual and physical testbeds presented the same behavior during this attack.

The second MiTM attack performed command injection and response injection in order to change the operation of the OpenPLC and change the data displayed on the HMI. The attack injected a command to increase the current pressure minimum and maximum setpoints on the OpenPLC by 13 PSI. This caused the pipeline to operate in a higher pressure than the original configuration. The attack also decreased 13 PSI from the response data sent to the HMI so that it would still receive the pressure

values according to the previously designated parameters. A comparison between the physical and virtual testbeds behavior for this attack is displayed in Figure. 4. As expected, both systems have a very similar response. When the MiTM attack is started on the physical systems a pressure drop is observed. This pressure drop is observed in the virtual system as well. This response is particularly important as it can be used as a signature while detecting similar MiTM attacks. After the dip in pressure the pump turns ON and builds up the necessary pressure. This is response is exactly reflected in the virtual system as well. Figure bellows shows the similar response of the physical and the virtual system.

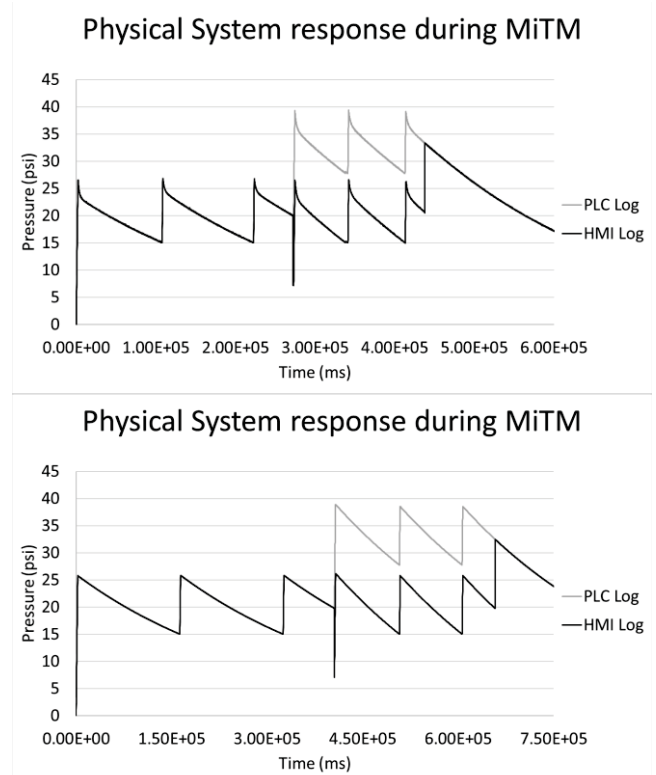


Figure 4. System response during MiTM attack

6. FUTURE WORK

It is important to be able to create accurate models of a virtual testbed as compared to the physical testbed. Therefore, future work would be to improve any discrepancies between these testbeds. The Hping attack had less response on the virtual testbed, in that it was not able to overwhelm the virtual machine by flooding the PLC with packets. Hence a virtual testbed running on an emulated hardware that is similar to the UniPi would be a good addition to the fidelity of the virtual testbed. Secondly, the system characteristics of the virtual and the physical controller can be compared to a standard PLC like SIEMENS S7 1200 controlling the same system. Thirdly, our study shows that the virtual testbed provides the necessary fidelity with the physical system to mimic similar response in certain scenarios. Hence the smaller virtual model can be scaled up to better represent a practical size cyber physical system.

7. CONCLUSION

Research in Industrial Control Systems (ICS) security is very essential in today's society, where there have now been recent attacks against SCADA systems. Hence cyber security researchers are looking for data on which further research can be performed.

Since full size cyber physical testbeds cannot be equipped in a laboratory environment, researchers tend to virtualize practical size ICS systems. Hence the question about fidelity of virtualized environment is eminent in any research of cyber physical system involving virtualized ICS. The virtual testbed need not only to offer highly accurate results but it has to offer sufficient high fidelity to mimic the response of different cyber-attacks and normal system failure modes. This work provides an elaborate comparison of a typical cyber physical testbed with a virtual counterpart and answers the big question of how much a virtual testbed can be trusted.

The virtual testbed being analyzed reflected high fidelity with the physical system other than the volumetric DoS attack. The volumetric DoS did not overwhelm the virtual testbed which was running on a much capable processor. Hence if a similar virtualized testbed is to be used to study the response due to DoS attack, the testbed has to be implemented on emulated hardware similar to the physical system.

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